

Towards Removing Assumptions in Class Incremental Learning

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Abstract

Continual learning in real-world applications must operate under privacy constraints, naturally imbalanced data, and reliance on pre-trained models, conditions where existing methods fail when these factors coexist. We introduce **Assumption-Free Class Incremental Learning (AFCIL)**, a realistic paradigm that simultaneously removes these limiting assumptions. To tackle AFCIL, we propose **Dynamic Expert Saturation (DES)**, a Mixture-of-Experts framework that leverages the natural head–tail hierarchy in data. DES progressively saturates expert capacity through two phases: first, training all experts on imbalanced data to capture dominant head-class patterns; then, freezing part of the capacity and fine-tuning the remaining experts with balanced sampling to specialize in tail classes without overwriting prior knowledge. Implemented with lightweight adapters over a frozen CLIP backbone, DES efficiently combines head and tail expertise through dynamic weighting. Under the AFCIL protocol, DES achieves state-of-the-art performance on CIFAR100-LT and ImageNet-LT, surpassing prior methods by up to 8.08% and 2.56%, respectively, while operating under the most realistic and difficult condition.

Introduction

Continual learning systems that can operate in real-world deployment scenarios are essential for autonomous vehicles adapting to new road conditions, medical AI systems learning from evolving patient populations, and industrial monitoring systems handling changing operational environments. These applications demand systems that learn continuously without storing previous data samples (for privacy compliance), handle naturally imbalanced data distributions, and leverage existing pre-trained infrastructure, yet no current method addresses all three constraints simultaneously.

Current continual learning methods fail catastrophically when deployed in realistic scenarios because they are fundamentally designed around unrealistic assumptions that make them unsuitable for practical deployment. The continual learning community has made impressive advances by tackling constraints in isolation: Long-Tailed Class Incremental Learning (LTCIL) (Liu et al. 2022; He 2024) handles imbalanced data, Exemplar-Free Class Incremental Learning (EFCIL) (Gomez-Villa et al. 2024) addresses

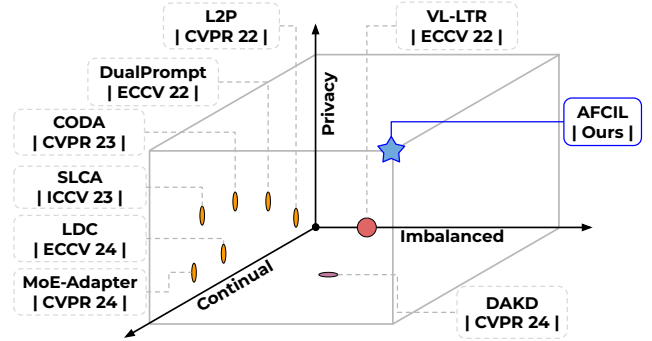


Figure 1: Positioning of our proposed DES against existing continual learning methods. Previous approaches tackle only one challenge at a time, such as imbalance (DAKD), privacy (LDC), or pre-trained weights (MoE, CODA, DualPrompt) and fail when these challenges overlap. In contrast, DES jointly addresses privacy and class imbalance, delivering robust performance in realistic continual learning settings.

exemplar-free learning but these setups restrict the use of pre-trained weights in contrast to mainstream continual learning (Wang et al. 2022a,b; Thengane et al. 2022; Smith et al. 2023; Zhang et al. 2023; Yu et al. 2024). Each shows great performance in their specific setup, creating an illusion of progress. However, when these assumptions are removed simultaneously, even state-of-the-art methods like MoE-Adapter (Yu et al. 2024), SLCA (Zhang et al. 2023), and CODA (Smith et al. 2023) suffer dramatic performance drops from >90% accuracy in controlled settings to 70-80% under realistic scenario. This creates a fundamental mismatch between research progress and practical reality: despite years of research and seemingly impressive results, practitioners still cannot deploy continual learning systems because privacy regulations prevent exemplar storage, data naturally arrives imbalanced, and organizations want to leverage existing pre-trained models, that is, all three constraints exist together, not in isolation.

Previous work on continual learning addresses these challenges through fragmented approaches that optimize for laboratory conditions rather than deployment realities. Exemplar-free methods like L2P (Wang et al. 2022b) and

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DualPrompt (Wang et al. 2022a) eliminate the need for storing previous samples but assume balanced data distributions and architectural flexibility. Long-tailed approaches such as DAKD (He 2024) handle imbalanced class distributions but still rely on exemplar storage. Mainstream continual learning techniques like SLCA (Zhang et al. 2023) and MoE-Adapter (Yu et al. 2024) leverage pre-trained vision-language models but struggle when data becomes highly imbalanced. Each approach tackles one dimension of the realistic constraint space while maintaining unrealistic assumptions about the others, creating a fragmented landscape where no single method handles realistic deployment conditions.

Our key insight is that realistic continual learning constraints should be embraced as natural features of the learning process rather than obstacles to be overcome. While previous work treats exemplar-free learning, data imbalance, and pre-trained requirements as separate problems requiring individual workarounds, we observe that these constraints actually reflect the fundamental structure of how knowledge naturally arrives and should be learned. Specifically, the natural head-tail hierarchy in imbalanced data distributions provides a road map for progressive expert specialization—first capturing the dominant patterns that define the knowledge space, then incrementally saturating remaining capacity with specialized knowledge for rare cases. This insight transforms the problem from fighting against realistic constraints through artificial balancing and data storage to strategically exploiting the natural learning hierarchy that these constraints reveal.

We introduce **Assumption-Free Class Incremental Learning (AFCIL)**, a unified continual learning paradigm that simultaneously removes three major assumptions—access to exemplars, class balance, and prevention of using pre-trained models, enabling realistic evaluation under practical deployment constraints as shown in Figure 1. To address this challenging setup, we propose **Dynamic Expert Saturation (DES)**, a novel Mixture-of-Experts framework that strategically exploits natural data hierarchies by first training all experts on imbalanced distributions to capture head class features, then selectively freezing experts while fine-tuning others with balanced sampling for tail class specialization. Our approach implements this through lightweight adapters over a frozen CLIP backbone, creating a principled expert allocation protocol with dynamic weighting mechanisms that effectively combines head and tail expert knowledge while maintaining computational efficiency.

We evaluate DES on CIFAR100-LT and ImageNet-LT across multiple imbalance ratios under the AFCIL protocol and demonstrate that it achieves state-of-the-art performance: 88.24% on CIFAR100-LT and 95.14% on ImageNet-LT, while operating under the most realistic scenario. These results represent improvements of **+8.08%** and **+2.56%** respectively over existing methods. By removing artificial constraints and designing for reality, we show that unified solutions can outperform isolated approaches, opening the door for practical deployment of continual learning systems in privacy-sensitive, pre-trained environments where data naturally arrives imbalanced.

Our contributions can be summarized as follows:

- We introduce AFCIL, a continual learning paradigm that jointly removes three unrealistic assumptions: exemplar access, class balance, and prevention of pre-trained models, creating a realistic testbed for future research.
- We propose DES, a Mixture-of-Experts architecture that leverages natural data hierarchies, using expert freezing and adapter-based fine-tuning to specialize for tail classes while retaining head class performance.
- Under the realistic AFCIL protocol, our method outperforms existing approaches by significant margins: **+8.08%** on CIFAR100-LT and **+2.56%** on ImageNet-LT.

Related Works

The evolution of continual learning research reveals a fundamental tension between achieving impressive laboratory results and addressing the messy constraints of real-world deployment. While the field has made substantial progress, this advancement has occurred through a fragmented approach where researchers tackle individual challenges in isolation, creating methods that excel under specific assumptions but fail when realistic constraints are combined. We study the different fragments as follows: **Exemplar-Free Learning: Addressing Privacy Constraints.** Experience Replay approaches store raw samples and replay them during new task learning. But these approaches cannot operate in privacy-sensitive domains like healthcare or finance where storing previous data violates regulatory requirements. In continual learning, many recent methods have focused on adapting large pre-trained models to new tasks. However, LDC(Gomez-Villa et al. 2024) takes a different approach by focusing on learning from scratch without relying on a pre-trained backbone. LDC(Gomez-Villa et al. 2024) introduces a novel method that tackles the problem of “semantic drift,” where the learned representations for old classes degrade as the model learns new ones. The proposed module is a lightweight component that predicts and corrects for these drifts in class prototypes, ensuring stable representation learning without the need for any stored exemplars or a fixed pre-trained foundation. By not building upon this foundational knowledge, the method misses a crucial opportunity to leverage the rich, general representations already learned from vast amounts of data. This choice not only makes the model less efficient, as it must learn representations from scratch for each task, but also potentially limits its performance compared to methods that can effectively adapt the powerful knowledge of a pre-trained model. **Continual Learning methods that use pretrained models: Addressing the usage of pretrained models.** Recent continual learning approaches, such as L2P(Wang et al. 2022b) and DualPrompt(Wang et al. 2022a), leverage unique prompt-based finetuning to adapt frozen pre-trained models to new tasks. This strategy effectively addresses privacy and storage concerns by learning task-specific prompts without storing previous data. L2P(Wang et al. 2022b) uses a prompt pool selected based on input, while DualPrompt(Wang et al. 2022a) utilizes complementary prompts

for general and task-specific knowledge. Similarly, CODA-Prompt(Smith et al. 2023) decomposes a Vision Transformer’s attention into general and task-specific prompts. SLCA(Zhang et al. 2023) focuses on preserving the pre-trained backbone’s knowledge with a ”Slow Learner” and aligning classifiers to prevent overfitting. MoE-Adapter(Yu et al. 2024) leverages Mixture-of-Experts (MoE) architectures to specialize different adapters for new tasks, demonstrating an efficient way to apply multimodal pre-trained knowledge to continual learning. While these methods make use of the extensive pretrained knowledge, they are primarily designed for balanced data, a critical limitation given the inherent long-tailed distributions of real-world datasets.

Long-Tailed Learning: Confronting Data Imbalance. In classification, a major challenge is long-tailed data distributions, where a few head classes dominate while many tail classes are under-represented. To counter the bias towards majority classes, Class-Balanced loss functions have become a key strategy. These methods re-weight the loss during training, giving higher importance to minority classes. While early versions used simple inverse frequency, more advanced techniques, like the ”effective number of samples” proposed by (Cui et al. 2019), offer a more nuanced approach. This re-weighting encourages models to learn better features for under-represented categories, improving overall performance on imbalanced datasets. Methods that handle imbalanced data are effective, but they are not designed for real-time information that arrives in a continuous stream. As a result, when these methods learn from new data, they tend to forget what they’ve already learned, leading to catastrophic forgetting

Long-Tailed Continual Learning: Confronting Data Imbalance Acknowledging that real-world data is inherently imbalanced, specialized methods like DAKD(He 2024) have emerged for long-tailed continual learning. DAKD(He 2024) tackles class imbalance by using dynamic knowledge distillation that adapts to shifting class frequencies across tasks, effectively improving performance on tail classes. However, most of these long-tailed continual learning methods often rely on unrealistic assumptions, such as exemplar storage or architectural flexibility. While they advance the field, these approaches don’t fully address the complete set of constraints found in practical deployment scenarios.

The Fragmentation Problem: Isolated Solutions for Combined Challenges The evolution of continual learning research reveals a consistent pattern: each methodological advance addresses one dimension of the realistic constraint space while maintaining unrealistic assumptions about others. Exemplar-free methods and the methods that make use of pretrained models solve privacy concerns but assume balanced data. Long-tailed approaches handle imbalance but require storage or architectural flexibility. This fragmentation creates a critical gap between research progress and practical deployment. While each approach demonstrates impressive performance within its designed constraints, none addresses the complete realistic scenario where exemplar-free learning, data imbalance, and use of pretrained model requirements must be satisfied simultaneously. The field has optimized for laboratory conditions that rarely exist in prac-

tice, creating a house of cards that collapses when exposed to real-world complexity. Our work addresses this fragmentation by proposing Assumption-Free Class Incremental Learning (AFCIL), which removes all three major assumptions simultaneously, and Dynamic Expert Saturation (DES), which is designed from the ground up for the combined realistic setting rather than adapted from assumption-dependent approaches. By embracing rather than fighting against realistic constraints, we demonstrate that unified solutions can actually outperform isolated approaches, opening the path toward truly deployable continual learning systems.

Methodology

Preliminaries

Problem Formulation Real-world continual learning is shaped by three factors often studied in isolation: (1) privacy regulations prohibiting storage of previous task data, (2) naturally imbalanced class distributions, and (3) reliance on pre-trained models rather than full fine-tuning. We introduce Assumption-Free Class Incremental Learning (AFCIL), the first paradigm that jointly incorporates these properties, creating an evaluation setting that mirrors realistic deployment.

AFCIL Setup The AFCIL setting models continual learning under conditions that closely resemble real data. We consider a sequence of tasks $\mathcal{T} = \{T_1, T_2, \dots, T_N\}$, where each task T_i introduces new classes $\mathcal{C}_i$. AFCIL preserves a global long-tailed distribution across all tasks. The total class set $\mathcal{C} = \bigcup_i \mathcal{C}_i$ is ordered by decreasing frequency, with the number of samples per class $c_j \in \mathcal{C}$ following:

$$N_{c_j} = N_{\max} \cdot \mu^j, \quad j \in \{0, 1, \dots, |\mathcal{C}| - 1\}, \quad (1)$$

where $N_{\max}$ is the size of the head class and $\mu \in (0, 1)$ controls the decay rate. The imbalance factor ρ is the ratio of the largest to smallest class sizes. This setup naturally induces a head–tail hierarchy that mirrors real-world distributions.

AFCIL removes common experimental assumptions and instead adopts conditions arising in practice:

- **Privacy-aware learning:** No data from previous tasks is stored ($\mathcal{D}_{t_{i-1}} = \emptyset$).
- **Naturally imbalanced data:** Class frequencies follow Eq. 1, reflecting real-world occurrence rates.
- **Pre-trained model reuse:** Learning builds on a frozen backbone (θ_{backbone} fixed), aligning with practical deployment pipelines.

This setting exposes a gap: as shown in our experiments in Table 1, methods designed for Exemplar-Free Class Incremental Learning (EFCIL) or Long-Tailed Class Incremental Learning (LTCIL) alone degrade severely when these properties coexist. This motivates the need for approaches that embrace, rather than counteract, the structure of real data.

Proposed Methodology: Dynamic Expert Saturation (DES)

We propose **DES**, a two-phase framework designed to exploit, rather than fight against, the natural head–tail hierar-

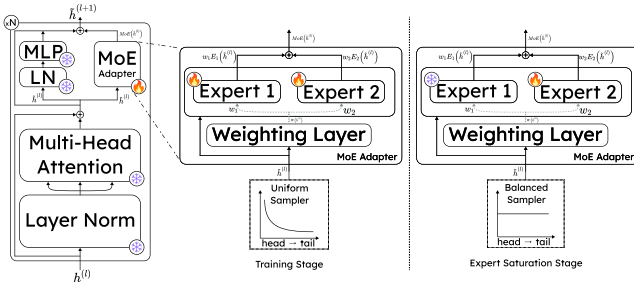


Figure 2: Architecture of **DES**. Each Transformer block has an MoE adapter with two experts. In the **Training Stage** (left), all experts (fire) train on imbalanced data using a uniform sampler, learning strong head-class features. In the **Expert Saturation Stage** (right), one expert is frozen (snowflake) to retain head knowledge while the other (fire) is fine-tuned on a balanced sampler to capture tail patterns. The Weighting Layer dynamically combines expert outputs in both stages.

chy in AFCIL. Existing methods fail because they treat privacy constraints, imbalance, and model freezing as obstacles; in contrast, DES leverages these as structural guides for learning.

The core idea is to build strong head-class representations first and then specialize experts for tail classes in a controlled manner. This staged progression mirrors how human expertise evolves—developing broad proficiency before refining rare-case mastery.

Architecture Design DES follows the MoE design of (Yu et al. 2024) while adapting it for AFCIL. Each Transformer block of the frozen CLIP backbone is augmented with K ($= 2$) lightweight expert adapters. The l -th block processes input $h^{(l)}$ through standard multi-head attention (MHA) and residual connections:

$$\tilde{h}^{(l)} = h^{(l)} + \text{MHA}(\text{LN}(h^{(l)})). \quad (2)$$

The output $\tilde{h}^{(l)}$ is then fed in parallel through the original MLP and the MoE adapter:

$$h^{(l+1)} = \tilde{h}^{(l)} + \text{MLP}(\text{LN}(\tilde{h}^{(l)})) + \text{MoE}(\text{LN}(\tilde{h}^{(l)})). \quad (3)$$

The MoE adapter combines expert outputs via a gating function:

$$\text{MoE}(\tilde{h}^{(l)}) = \sum_{k=1}^K w_k E_k(\tilde{h}^{(l)}), \quad (4)$$

where the gating weights are computed by a linear projection:

$$[w_1, w_2, \dots, w_K] = W(\tilde{h}^{(l)}), \quad (5)$$

with $W \in \mathbb{R}^{K \times d}$ being the gating weight matrix that maps the input feature of dimension d to K expert weights. Each expert $E_k(\cdot)$ is a lightweight MLP:

$$E_k(\tilde{h}^{(l)}) = \text{MLP}_k(\tilde{h}^{(l)}). \quad (6)$$

This integration enables expert-driven refinement while preserving the frozen backbone’s strong representations.

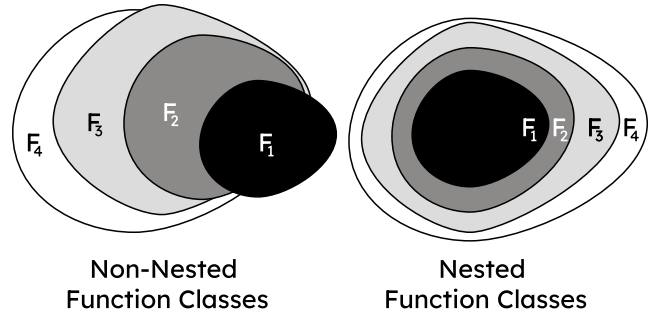


Figure 3: Illustration of **nested** vs. **non-nested** function classes. In non-nested classes (left), adding new functions can disrupt previously learned representations, causing interference and catastrophic forgetting. In nested classes (right), each function space F_{i+1} extends F_i while preserving its knowledge. This principle underlies DES, where frozen experts maintain strong head-class representations, and active experts progressively learn residual refinements for tail classes without overwriting prior knowledge.

Training Stage. This stage as shown in Fig. 2, builds robust features using the naturally imbalanced dataset $\mathcal{D}_{\text{imb}}^{t_i}$. Since head classes dominate, this stage is effectively realized through standard cross-entropy training and uniform sampler, identical to the initial training strategy in (Yu et al. 2024). All experts are trained jointly within the frozen backbone, allowing them to internalize the patterns of frequent classes. The objective is:

$$\mathcal{L}_1 = - \sum_{(\mathbf{x}, y) \in \mathcal{D}_{\text{imb}}^{t_i}} \log P(y|\mathbf{x}), \quad (7)$$

where class probabilities follow the CLIP paradigm:

$$P(y = j|\mathbf{x}) = \frac{\exp(\cos(F_I(\mathbf{x}), F_T(c_j)))}{\sum_{j'} \exp(\cos(F_I(\mathbf{x}), F_T(c_{j'})))}. \quad (8)$$

Here, $F_I(\mathbf{x})$ is the image embedding enriched by both MLP and MoE pathways, and $F_T(c_j)$ is the class text embedding.

Expert Saturation Stage. This stage of DES as shown in Fig. 2, builds on the observation that once a reliable function for head classes has been established, additional capacity should refine rather than relearn it. To achieve this, we treat the outputs of the frozen experts as a fixed base function ($f_{\text{frozen}}(\mathbf{x})$), which remains unchanged during this stage and encapsulates stable head-class knowledge. The active experts then learn a residual refinement ($\Delta f_{\text{tail}}(\mathbf{x})$) that focuses exclusively on underrepresented tail classes.

This setup can be viewed as operating in a nested function space $\mathcal{F}_{\text{nested}} \supseteq \mathcal{F}_{\text{frozen}}$, where each additional component extends the previously learned function without disrupting it. As illustrated in Fig. 3, such nesting avoids the instability typical of non-nested function classes and mirrors the principle of residual learning (He et al. 2016), where new components capture only what is missing.

Concretely, we randomly choose $M < K$ experts to freeze as head-class specialists,

$$\mathcal{E}_{\text{frozen}} \subset \{E_1, \dots, E_K\}, \quad |\mathcal{E}_{\text{frozen}}| = M,$$

while the remaining active experts

$$\mathcal{E}_{\text{active}} = \{E_1, \dots, E_K\} \setminus \mathcal{E}_{\text{frozen}}$$

are fine-tuned on a balanced sampler $\mathcal{D}_{\text{bal}}$ to model $\Delta f_{\text{tail}}(\mathbf{x})$. The final feature representation is expressed as:

$$F(\mathbf{x}) = f_{\text{frozen}}(\mathbf{x}) + \Delta f_{\text{tail}}(\mathbf{x}), \quad (9)$$

where the residual term learns to capture tail-specific variations while leaving the head-class function intact.

Following the CLIP paradigm, class probabilities are computed via cosine similarity between the refined feature $F(\mathbf{x})$ and class text embeddings $F_T(c_j)$ (as in Eq. 8). The training objective for this stage is:

$$\mathcal{L}_2 = - \sum_{(\mathbf{x}, y) \in \mathcal{D}_{\text{bal}}} \log P(y|\mathbf{x}). \quad (10)$$

This training strategy not only preserves the strong representations learned in the initial stage but also channels new capacity towards capturing the rare patterns of tail classes, achieving specialization without catastrophic interference. Much like the way human expertise develops—where professionals first master common cases before honing skills on rare but critical edge cases—this progression strengthens performance on frequent scenarios while cultivating specialized competency for exceptional ones. The complete DES training procedure is designed to be both principled and practical. Algorithm 1 summarizes the key steps.

Experiments

Experimental Setup

We conduct extensive experiments to validate our DES approach within the proposed AFCIL setting. Our evaluation reveals that while existing state-of-the-art continual learning methods excel in their tailored scenarios, their performance deteriorates sharply when all realistic deployment conditions are applied together. Through thorough ablation studies and in-depth analysis, we demonstrate that DES effectively bridges this gap, achieving superior results by strategically exploiting the inherent hierarchies present in the data. We use the AdamW optimizer with a learning rate of 10^{-3} , a cosine scheduler with linear warmup (10% of total training steps), and apply the same training setup across all experiments. To ensure simplicity and efficiency, we fix the number of experts to $K = 2$. Since at least one expert must remain active for effective mixture saturation, we set $M = 1$.

Datasets. To thoroughly evaluate the effectiveness of our approach in realistic continual learning scenarios, we conduct experiments on two widely-used benchmarks, adapted to the challenging AFCIL setting. Our design enforces real-world deployment constraints, ensures fair comparison across methods, and strictly follows an exemplar-free protocol where no past data is stored. Following established protocols from prior LTCIL work (Liu et al. 2022), we adopt a shuffled task setup, which better reflects realistic scenario. The total class set is divided equally among T tasks, and we evaluate two configurations: $T = 5$ and $T = 10$. For example, with 100 classes, the 5-task setup allocates 20 classes

Algorithm 1: Dynamic Expert Saturation (DES) for Task_t

Require: Imbalanced dataset $\mathcal{D}_{\text{imb}}$, number of experts K , number of frozen experts M , training epochs E_1, E_2
1: **Initialize:** Mixture-of-Experts (MoE) adapters $\{E_1, \dots, E_K\}$ with gating network W producing weights w_k .

2: **Training Stage (All Experts Train)**

3: **for** epoch = 1 $\rightarrow$ E_1 **do**

4: Sample a mini-batch from $\mathcal{D}_{\text{imb}}$ (natural long-tailed distribution).

5: Forward pass through all experts:

$$\text{MoE}(\tilde{h}^{(l)}) = \sum_{k=1}^K w_k E_k(\tilde{h}^{(l)})$$

6: Compute predictions using the weighted ensemble and loss $\mathcal{L}_1$.

7: Update parameters of all experts $\{E_1, \dots, E_K\}$ with Adam.

8: **end for**

9: **Expert Saturation Stage (Partial Training)**

10: Randomly choose M experts to freeze: $\mathcal{E}_{\text{frozen}} \subset \{E_1, \dots, E_K\}$.

11: Define active experts: $\mathcal{E}_{\text{active}} = \{E_1, \dots, E_K\} \setminus \mathcal{E}_{\text{frozen}}$.

12: Construct a balanced dataset $\mathcal{D}_{\text{bal}}$ with equal samples per class.

13: **for** epoch = 1 $\rightarrow$ E_2 **do**

14: Sample a mini-batch from $\mathcal{D}_{\text{bal}}$.

15: Forward pass through *all* experts using the gating-weighted combination:

$$\text{MoE}(\tilde{h}^{(l)}) = \sum_{k=1}^K w_k E_k(\tilde{h}^{(l)})$$

16: Compute predictions and loss $\mathcal{L}_2$.

17: Update only parameters of $\mathcal{E}_{\text{active}}$ (frozen experts remain unchanged).

18: **end for**

per task, while the 10-task setup uses 10 classes per task. We use CIFAR100-LT and ImageNet-LT, both containing 100 classes with long-tailed distributions characterized by an imbalance factor $\rho = \frac{N_{\text{max}}}{N_{\text{min}}}$, where N_{max} and N_{min} denote the maximum and minimum class frequencies, respectively. We evaluate across imbalance levels $\rho = 100, 50, 10$ as in (Liu et al. 2022). For $\rho = 100$, CIFAR100-LT ranges from 500 samples in the head class to 5 in the tail class, while ImageNet-Subset ranges from 1,300 to 13 samples.

AFCIL Evaluation Protocol. At the core of our experimental design is the AFCIL protocol, which embraces a setting free from restrictive assumptions, aligning closely with real-world continual learning. First, AFCIL operates without reliance on stored exemplars, naturally accommodating privacy-sensitive applications where retaining past data is neither possible nor necessary. Second, it learns directly from naturally imbalanced data streams without artificially

	CIFAR100-LT		ImageNet-LT	
	5 Task	10 Task	5 Task	10 Task
DAKD (CVPR24)	31.26	21.33	35.50	23.61
LDC (ECCV24)	37.17	28.39	63.84	59.44
Continual CLIP	79.36	80.16	89.39	89.82
L2P (CVPR22)	69.97	70.09	88.23	87.93
DualPrompt (ECCV22)	70.70	71.11	89.82	89.65
CODA (CVPR23)	82.33	79.19	90.14	90.25
SLCA (ICCV23)	72.25	73.13	92.75	92.58
MoE-Adapter (CVPR24)	77.65	77.78	88.71	87.39
DES (Ours)	87.65	88.24	94.00	95.14
	5.32 $\uparrow$	8.08 $\uparrow$	1.25 $\uparrow$	2.56 $\uparrow$

Table 1: Accuracies by category (LTCIL, EFCIL, CLIP+Continual Learning) with imbalance ratio $\rho = 100$ for 5-task and 10-task setups on CIFAR100-LT and ImageNet-LT. **Best** and second-best scores are highlighted, with improvements marked by $\uparrow$. DES consistently outperforms all baselines.

enforcing balance, thereby capturing the authentic challenges posed by skewed distributions. Third, it builds on top of a fixed pre-trained CLIP backbone, leveraging existing models without requiring costly retraining. Together, these properties ensure that AFCIL reflects the unconstrained and practical nature of continual learning in deployment

Comparison Methods. We benchmark our method against a diverse set of continual learning approaches represented in the table. Specifically, we include: (1) Long-Tailed Class Incremental Learning (LTCIL) methods including DAKD (He 2024), (2) Exemplar-Free Class Incremental Learning (EFCIL) methods like LDC (Gomez-Villa et al. 2024), and (3) for Continual + CLIP approaches, we evaluate Continual CLIP (Thengane et al. 2022), L2P (Wang et al. 2022b), DualPrompt (Wang et al. 2022a), CODA (Smith et al. 2023), SLCA (Zhang et al. 2023) and MoE-Adapter (Yu et al. 2024).

Results & Discussions

We evaluate DES on CIFAR100-LT and ImageNet-LT under multiple imbalance ratios ($\rho = 100, 50, 10$) for both 5-task and 10-task configurations. Tables 1–3 present a comprehensive comparison against a diverse set of state-of-the-art methods, including prompt-based approaches (L2P (Wang et al. 2022b), DualPrompt (Wang et al. 2022a)), contrastive strategies (CODA (Smith et al. 2023)), and mixture-of-experts designs (MoE-Adapter (Yu et al. 2024)). Across all experimental settings, DES demonstrates strong or superior performance, with distinct trends emerging across imbalance levels. Under severe imbalance ($\rho = 100$), where tail classes are extremely underrepresented, DES achieves its largest gains, outperforming the strongest baselines by up to **8.08%** on CIFAR100-LT and **2.56%** on ImageNet-LT. These improvements are especially significant over long-tail specific methods like SLCA and CODA, indicating that DES effectively mitigates head-tail volatility and enhances tail-class retention without sacrificing head-class accuracy.

	CIFAR100-LT		ImageNet-LT	
	5 Task	10 Task	5 Task	10 Task
DAKD (CVPR24)	24.89	17.35	30.66	20.97
LDC (ECCV24)	39.43	27.90	66.11	54.89
Continual CLIP	79.36	80.16	89.39	89.82
L2P (CVPR22)	74.35	74.75	89.40	88.87
DualPrompt (ECCV22)	75.30	76.03	91.22	91.29
CODA (CVPR23)	<u>86.02</u>	<u>83.77</u>	91.52	90.98
SLCA (ICCV23)	75.89	76.67	93.20	93.66
MoE-Adapter (CVPR24)	79.38	78.58	92.69	89.69
DES (Ours)	87.76	89.33	95.22	95.38
	1.74 $\uparrow$	5.56 $\uparrow$	2.02 $\uparrow$	1.72 $\uparrow$

Table 2: Accuracies by category (LTCIL, EFCIL, CLIP+Continual Learning) with imbalance ratio $\rho = 50$ for 5-task and 10-task setups on CIFAR100-LT and ImageNet-LT. **Best** and second-best scores are highlighted, with improvements marked by $\uparrow$. DES consistently outperforms all baselines.

	CIFAR100-LT		ImageNet-LT	
	5 Task	10 Task	5 Task	10 Task
DAKD (CVPR24)	31.26	21.33	35.50	23.61
LDC (ECCV24)	37.17	28.39	63.84	59.44
Continual CLIP	79.36	80.16	89.39	89.82
L2P (CVPR22)	80.09	79.67	91.75	91.74
DualPrompt (ECCV22)	80.47	80.45	93.06	93.24
CODA (CVPR23)	89.99	87.90	93.18	91.61
SLCA (ICCV23)	80.12	80.63	93.97	94.09
MoE-Adapter (CVPR24)	82.07	82.19	90.70	91.14
DES (Ours)	<u>89.51</u>	90.45	95.81	96.08
	0.48 $\downarrow$	2.55 $\uparrow$	1.84 $\uparrow$	1.99 $\uparrow$

Table 3: Accuracies by category (LTCIL, EFCIL, CLIP+Continual Learning) with imbalance ratio $\rho = 10$ for 5-task and 10-task setups on CIFAR100-LT and ImageNet-LT. **Best** and second-best scores are highlighted, with improvements marked by $\uparrow$ or $\downarrow$. DES achieves the highest performance on ImageNet-LT while showing competitive results on CIFAR100-LT.

For moderate imbalance ($\rho = 50$), DES continues to deliver high performance, achieving the best results on both datasets in all settings. It surpasses the next-best competitor by margins of up to **5.56%** on CIFAR100-LT and **2.02%** on ImageNet-LT.

When the imbalance ratio is small ($\rho = 10$), the performance landscape shifts. DES remains the top-performing approach on ImageNet-LT, achieving improvements of up to **1.99%** over the next best method, underscoring its scalability to large-scale datasets. However, in CIFAR100-LT, CODA achieves marginally higher accuracy, in the 5-task case. This indicates that DES, while designed for highly skewed distributions, does not over-specialize and stays competitive even in near-balanced scenarios.

Another key observation is the stability of DES across increasing task numbers. Many baselines, including CODA

Component Analysis		
Frozen Expert	Balanced Sampler	Accuracy
✗	✗	83.52
✓	✗	83.77
✗	✓	87.16
✓	✓	87.65
Expert Selection Strategy Analysis		
Expert Selection Strategy		Accuracy
Top-weighted (Local)		87.34
Top-weighted (Global)		87.45
Round Robin		87.52
Random		87.65
Saturation Technique Analysis		
Saturation Technique		Accuracy
Weighted Loss		83.92
Balanced Sampling		87.65

Table 4: Comprehensive ablation study for DES on CIFAR100-LT dataset. **Top:** Impact of different components. **Middle:** Analysis of expert selection strategies. **Bottom:** Comparison of saturation techniques. **Best** and second-best scores are highlighted with improvements marked by $\uparrow$.

and MoE-Adapter, exhibit more pronounced drops when transitioning from 5-task to 10-task setups. In contrast, DES maintains stable performance across both splits, suggesting effective reuse of learned representations and reduced catastrophic forgetting. Finally, comparing DES to prompt-based methods (L2P, DualPrompt) and existing mixture-of-experts designs reveals why it excels: prompt-based methods improve task adaptation but fail to address long-tail dynamics, and existing MoE approaches lack targeted mechanisms for balancing head and tail learning. DES unifies stable representation learning with explicit tail-aware specialization, enabling it to consistently outperform or match state-of-the-art methods across varying imbalance ratios and task configurations. Its most substantial benefits emerge in challenging long-tailed settings, validating its effectiveness in addressing the core challenges of AFCIL.

Ablation Studies

We perform ablations on CIFAR100-LT to evaluate the role of DES components and validate design choices. Table 4 reports results for three factors: component contributions, expert selection strategies, and saturation techniques.

Component analysis. The top section of Table 4 evaluates two key elements: freezing one expert and using a balanced sampler in the second phase. Removing both yields the lowest accuracy (83.52%), confirming their importance. Freezing only (✓) moderately improves performance (83.77%) by preserving head knowledge but lacks tail emphasis. Using only a balanced sampler (✓) performs better (87.16%), showing strong benefits from tail rebalancing. Combining both (✓✓) achieves the best result (87.65%), indicating that head preservation and tail balancing complement each other.

Expert selection. The middle section compares four strategies for choosing which expert to freeze: (1) *Round Robin*, where freezing rotates across experts with each task; (2) *Top-weighted (Local)*, where the expert with the highest gating weight in each block is frozen; (3) *Top-weighted (Global)*, where the expert with the highest average weight across all blocks is frozen; and (4) *Random*, where an expert is frozen without preference. Surprisingly, random selection achieves the best accuracy (87.65%), slightly surpassing Round Robin (87.52%). The weight-based strategies (local 87.34%, global 87.45%) perform marginally worse, despite appearing more informed. This suggests that gating weights, dominated by head classes in the first stage, bias the frozen expert toward heads and hinder tail specialization. Random and Round Robin strategies avoid this bias, enabling the active expert to specialize more effectively on tails.

Saturation technique. The bottom section analyzes methods to emphasize tail classes during specialization. We compare a *Weighted Loss* (Liu et al. 2022) against a *Balanced Sampling*. While both enhance tail learning, the balanced sampler significantly outperforms weighted loss (87.65% vs. 83.92%). Directly controlling the input distribution proves more effective than scaling the loss, ensuring consistent tail exposure without destabilizing head representations.

Overall, the ablation study shows that freezing plus balanced sampling is critical for optimal performance. Avoiding head-biased selection improves specialization, and the balanced sampler is the most effective saturation technique for tail-focused learning.

Conclusion and Future Work

This paper introduces **Assumption-Free Class Incremental Learning (AFCIL)** paradigm, addressing the gap between laboratory-designed continual learning methods and the constraints of real-world deployment. By removing assumptions about exemplar storage, data balance, and architectural flexibility, AFCIL establishes a realistic evaluation setting. To tackle this setting we propose **Dynamic Expert Saturation (DES)** which leverages a Mixture-of-Experts design: first capturing head-class knowledge on imbalanced data, then progressively specializing for tail classes by freezing experts and fine-tuning with a tail-favoring saturation process. Our experiments show that DES consistently outperforms state-of-the-art baselines across multiple imbalance ratios and task splits, with the largest gains observed in scenarios where current methods typically fail. Looking ahead, AFCIL, alongside DES, opens several promising directions. This work lays the groundwork for methods that leverage, rather than circumvent, real-world constraints.

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